CareLoop SG · Powered by Qloo Taste AI

Quiet, culturally grounded support for seniors with high blood pressure, between doctor visits

THE PROBLEM

Most of hypertension care happens at home, between visits. Readings drift, pills get missed, a senior feels unwell or isolated and says nothing. Alert apps fire fixed "BP over 140" alarms so often that families stop listening — while generic AI companions suggest generic salads, loud gyms, or unsafe foods that ignore a senior's cultural roots, medications, and the day's weather.

WHAT CARELOOP DOES

A short daily check-in, by voice or big buttons, in English, Singlish, Mandarin, Malay or Tamil. CareLoop learns **each senior's own normal**, stays quiet while things are normal, and steps up gently when patterns persist. Between check-ins, **Qloo Taste AI** turns a senior's cultural memories (e.g. Teresa Teng, Teochew porridge, 1960s Shaw Brothers films, P. Ramlee) into safe, nearby meals, nostalgic playlists, and Active Ageing Centre programmes — filtered through deterministic clinical, pharmacist-leaflet, and live Singapore haze rules.

WHO USES IT

- **Seniors:** a large-text app with voice check-ins, plain-word medicine guides, Qloo-matched daily activities, and one-tap memory erasure.
- **Family:** quiet trends, evidence-backed handovers, a Qloo **Family Visit Kit** (what to tap/bring on Sunday), and a one-page GP brief.
- **Centre staff:** a caseload console sorted by urgency, plus a **Multi-Senior Qloo Group Intersection** & **Kopi Kakis** buddy matcher.

HOW QLOO TASTE AI + CLINICAL RULES WORK TOGETHER

- **1. Resolve (`qloo_search`):** maps cultural seeds to canonical Qloo entity IDs.
- **2. Cross-Domain (`qloo_insights`):** queries `urn:entity:place`, `artist`, `movie` with `signal.demographics.age=55_and_older` & Singapore WKT polygons.
- **3. Clinical Filter (`check_health_constraints`):** blocks high-sodium, gout purine, statin/amlodipine grapefruit-pomelo leaflet clashes, and outdoor walks when PSI > 100.
- **4. Guardrail (`guardrail_check`):** verifies bilingual wording in EN / ZH / MS / TA.

HOW IT USES AI, SAFELY

Rules decide; AI and Qloo handle language and cultural taste. A language model reads check-ins to catch what keyword rules miss ("my chest feel like got elephant sitting on top" → chest-pain red flag) and words gentle notes around Qloo's structured entity graph. Neither the LLM nor Qloo ever decides whether a reading is dangerous or overrides a medical/medication/haze filter. Every message passes `guardrail.py` (no diagnosis, no dose changes, no invented numbers) or falls back to a fixed template.

TESTED BLIND: CLINICAL EVALUATION & 38-PERSONA QLOO ABLATION

Evaluated on a frozen 720-day clinical holdout and a 38-persona multicultural Singapore senior benchmark (EN / ZH / MS / TA) comparing Qloo Taste AI + deterministic filters against an ungrounded LLM:

0.0%	**82.2 / 100**	**5 of 5 · 0**
clinical, leaflet or haze violations (scripted generic-LLM baseline: 100%)	cultural specificity with Qloo (generic LLM: 19.0 / 100)	red flags caught · 0 false alarms per senior-month

HOW IT'S BUILT

React + Vite web app · Python FastAPI backend · live Qloo Taste AI client (`hackathon.api.qloo.com` `/search` & `/v2/insights`) with deterministic offline fallback · single-source clinical rules (`thresholds.py`, `medication_knowledge.py`) · live NEA PSI haze & rainfall (`data.gov.sg`) · 183+ automated tests.

Live demo: careloop-sg-qloo.onrender.com · **Qloo Judge Console:** /pitch · **Senior app:** /senior?senior=tan · **Family + Activities:** /care/family · **Centre + Cohort:** /care/aac
Code: github.com/DrunkWiz/careloop-sg-qloo · **Demo Video:** careloop-qloo-launch-video.mp4 · **Submission:** docs/submission/qloo-submission.md · **Architecture:** docs/architecture.md · **Evaluation:** docs/evaluation.md (`GET /eval/taste`)

All people in the demo are synthetic. CareLoop is a prototype, not a medical device, and is not connected to any clinic system. Clinical thresholds, red-flag wording and medicine notes are placeholders until reviewed by a clinician. Author: Nathan.